Week 6 – Product Review and Rating Management System.

The e-commerce company aims to develop a system that allows customers to share their opinions about the products they purchase. After successfully completing an order, customers can submit ratings and written reviews based on their experience. The system maintains all customer feedback in a database and helps the company evaluate product quality and customer satisfaction. It also makes it easier to identify the most popular products and products with high ratings. To achieve this, students will develop a Product Review and Rating Module that is linked with the existing Customer and Product modules.

1. Create the Review table

```
SQL> CREATE TABLE REVIEW (
  2  REVIEWID VARCHAR2(7),
  3  CUSTOMERID NUMBER,
  4  PRODUCTID VARCHAR2(7),
  5  RATING NUMBER(1),
  6  REVIEWTEXT VARCHAR2(50),
  7  REVIEWDATE DATE,
  8  CONSTRAINT PK_REVIEW PRIMARY KEY (REVIEWID),
  9  CONSTRAINT FK_REVIEW_CUSTOMER FOREIGN KEY (CUSTOMERID)
 10  REFERENCES CUSTOMER(CUSTOMER_ID),
 11  CONSTRAINT FK_REVIEW_PRODUCT FOREIGN KEY (PRODUCTID)
 12  REFERENCES PRODUCT(PRODUCT_ID),
 13  CONSTRAINT CHK_RATE CHECK (RATING BETWEEN 1 AND 5)
 14  );

Table created.

SQL> DESC REVIEW;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 REVIEWID                                  NOT NULL VARCHAR2(7)
 CUSTOMERID                                         NUMBER
 PRODUCTID                                          VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEWTEXT                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

```
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0002', 'CUST00002', 'P000002', 4, 'GOOD PRODUCT', DATE '2026-02-10');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0003', 'CUST00003', 'P000003', 5, 'GREAT QUALITY', DATE '2026-03-05');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0004', 'CUST00004', 'P000004', 3, 'AVERAGE PRODUCT', DATE '2026-03-15');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0005', 'CUST00005', 'P000005', 4, 'GOOD PERFORMANCE', DATE '2026-04-12');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0006', 'CUST00006', 'P000006', 5, 'HIGHLY RECOMMENDED', DATE '2026-05-20');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0007', 'CUST00007', 'P000007', 2, 'POOR QUALITY', DATE '2026-06-08');

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  ('REV0008', 'CUST00008', 'P000008', 4, 'VALUE FOR MONEY', DATE '2026-06-15');

1 row created.
```

3. Display all reviews

```
SQL> SELECT * FROM Review;

REVIEWI CUSTOMERID PRODUCT     RATING REVIEWTEXT                                          REVIEWDAT
------- ---------- ------- ---------- -------------------------------------------------- ---------
REV0003          3 P000003          3 Average product                                    15-JAN-26
REV0004          1 P000004          5 Excellent performance                              18-JAN-26
REV0005          2 P000005          2 Quality needs improvement                          20-JAN-26
REV0009          3 P000003          1 Poor product quality                               05-FEB-26
REV0010          1 P000004          4 Worth purchasing                                   08-FEB-26
REV0011          2 P000005          5 Outstanding product                                10-FEB-26
REV0015          3 P000003          3 Decent product                                     20-FEB-26
REV0008          2 P100001          3 Satisfactory quality                               03-FEB-26
REV0012          3 P200002          2 Not satisfied                                      12-FEB-26
REV0013          1 P000003          4 Good performance                                   15-FEB-26
REV0014          2 P000004          5 Amazing quality                                    18-FEB-26
REV0016          1 P100001          5 Excellent quality                                  22-FEB-26
REV0017          2 P200002          4 Worth purchasing                                   25-FEB-26
REV0018          3 P000003          5 Highly recommended                                 28-FEB-26

14 rows selected.
```

ENSURE THIS BY ADDING CONSTRAINT

```
SQL> ALTER TABLE REVIEW ADD CONSTRAINT CHK_RATE CHECK(RATING >= 1 AND RATING <= 5);

Table altered.
```

```
SQL> INSERT INTO REVIEW VALUES('REV0019','CUST000001','P000001',6,'EXCELLENT PRODUCT',SYSDATE);
INSERT INTO REVIEW VALUES('REV0019','CUST000001','P000001',6,'EXCELLENT PRODUCT',SYSDATE)
*
ERROR at line 1:
ORA-02290: check constraint (C##ECOMMERCE.CHK_RATE) violated
```

3.UPDATE REVIEW COMMENTS

```
SQL> UPDATE REVIEW SET REVIEW_TEST = 'GOOD QUALITY BUT COULD BE IMPROVED' WHERE REVIEW_ID ='REV0013';

1 row updated.
```

4.REMOVE INAPPROPRIATE OR INVALID REVIEWS

```
SQL> DELETE FROM REVIEW WHERE REVIEW_ID = 'REV0015';

1 row deleted.
```

4.RETRIEVE PRODUCT REVIEW DETAILS

1.DISPLAY ALL REVIEWS FOR A PRODUCT

```
SQL> SELECT * FROM REVIEW WHERE PRODUCT_ID = 'P000012';

REVIEW_ID  CUSTOMER_I PRODUCT_ID     RATING REVIEW_TEST                                         REVIEW_DA
---------- ---------- ---------- ---------- -------------------------------------------------- ---------
REV0012    CUST00012  P000012             3 DECENT PRODUCT                                     18-AUG-26
```

2.DISPLAY CUSTOMER NAME WITH THEIR REVIEWS

```
SQL> SELECT
  2      C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3      R.ReviewID,
  4      R.ProductID,
  5      R.Rating,
  6      R.ReviewText
  7  FROM Customer C
  8  JOIN Review R
  9  ON C.Customer_ID = R.CustomerID;

CUSTOMER_NAME                                                                                      REVIEWI PRODUCT     RATING
-------------------------------------------------------------------------------------------------- ------- ------- ----------
REVIEWTEXT
--------------------------------------------------
Arun Kumar                                                                                         REV0016 P100001          5
Excellent quality

Arun Kumar                                                                                         REV0013 P000003          4
Good performance

Arun Kumar                                                                                         REV0010 P000004          4
Worth purchasing

Arun Kumar                                                                                         REV0004 P000004          5
Excellent performance

Meena Ravi                                                                                         REV0017 P200002          4
Worth purchasing

Meena Ravi                                                                                         REV0014 P000004          5
Amazing quality

Meena Ravi                                                                                         REV0008 P100001          3
Satisfactory quality

Meena Ravi                                                                                         REV0011 P000005          5
Outstanding product

Meena Ravi                                                                                         REV0005 P000005          2
Quality needs improvement

Karthik Raj                                                                                        REV0018 P000003          5
Highly recommended

Karthik Raj                                                                                        REV0012 P200002          2
Not satisfied

Karthik Raj                                                                                        REV0015 P000003          3
Decent product

Karthik Raj                                                                                        REV0009 P000003          1
Poor product quality

Karthik Raj                                                                                        REV0003 P000003          3
Average product


14 rows selected.
```

3.FIND PRODUCTS HAVING MAXIMUM REVIEWS

```
SQL> SELECT
  2       P.Product_ID,
  3       P.Product_Name,
  4       COUNT(R.ReviewID) AS Review_Count
  5  FROM Product P
  6  JOIN Review R
  7  ON P.Product_ID = R.ProductID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Review_Count DESC;

PRODUCT PRODUCT_NAME         REVIEW_COUNT
------- -------------------- ------------
P000003 Face Cream                      5
P000004 Lipstick                        3
P100001 Laptop                          2
P200002 Kids Shoes                      2
P000005 Face Wash                       2
```

4.DISPLAY RECENT CUSTOMER FEEDBACK

```
SQL> SELECT *
  2  FROM (
  3      SELECT
  4          C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  5          P.Product_Name,
  6          R.Rating,
  7          R.ReviewText AS Customer_Feedback,
  8          R.ReviewDate AS Review_Date
  9      FROM Review R
 10      JOIN Customer C
 11          ON R.CustomerID = C.Customer_ID
 12      JOIN Product P
 13          ON R.ProductID = P.Product_ID
 14      ORDER BY R.ReviewDate DESC
 15  )
 16  WHERE ROWNUM <= 5;

CUSTOMER_NAME                                                                                                    PRODUCT_NAME         RATING
---------------------------------------------------------------------------------------------------------------- -------------------- ----------
CUSTOMER_FEEDBACK                                 REVIEW_DA
------------------------------------------------- ---------
Karthik Raj                                                                                                      Face Cream                5
Highly recommended                                28-FEB-26

Meena Ravi                                                                                                       Kids Shoes                4
Worth purchasing                                  25-FEB-26

Arun Kumar                                                                                                       Laptop                    5
Excellent quality                                 22-FEB-26

Karthik Raj                                                                                                      Face Cream                3
Decent product                                    20-FEB-26

Meena Ravi                                                                                                       Lipstick                  5
Amazing quality                                   18-FEB-26
```

5.RETRIEVE REVIEWS WITH RATING ABOVE 5

```
SQL> SELECT *
  2  FROM Review
  3  WHERE Rating > 4;

REVIEWI CUSTOMERID PRODUCT     RATING REVIEWTEXT                                          REVIEWDAT
------- ---------- ------- ---------- -------------------------------------------------- ---------
REV0004          1 P000004          5 Excellent performance                              18-JAN-26
REV0011          2 P000005          5 Outstanding product                                10-FEB-26
REV0014          2 P000004          5 Amazing quality                                    18-FEB-26
REV0016          1 P100001          5 Excellent quality                                  22-FEB-26
REV0018          3 P000003          5 Highly recommended                                 28-FEB-26

SQL>
```

5.CALCULATE AVERAGE PRODUCT RATINGS

TASKS:

1.CALCULATE AVERAGE RATING FOR EACH PRODUCT

```
SQL> SELECT
  2      P.Product_Name,
  3      ROUND(AVG(R.Rating), 2) AS Average_Rating
  4  FROM Product P
  5  JOIN Review R
  6  ON P.Product_ID = R.ProductID
  7  GROUP BY P.Product_Name;

PRODUCT_NAME         AVERAGE_RATING
-------------------- --------------
Face Wash                       3.5
Lipstick                       4.67
Kids Shoes                        3
Laptop                            4
Face Cream                      3.2
```

2.COUNT THE NUMBER OF REVIEWS PER PRODUCT

```
SQL> SELECT
  2      P.Product_Name,
  3      COUNT(R.ReviewID) AS Count_Rating
  4  FROM Product P
  5  LEFT JOIN Review R
  6  ON P.Product_ID = R.ProductID
  7  GROUP BY P.Product_Name;

PRODUCT_NAME         COUNT_RATING
-------------------- ------------
Face Wash                       2
Lipstick                        3
Kids Shoes                      2
Laptop                          2
Face Cream                      5
```

3.FIND HIGHEST-RATED PRODUCTS.

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      ROUND(AVG(R.Rating), 2) AS Average_Rating
  5  FROM Product P
  6  JOIN Review R
  7  ON P.Product_ID = R.ProductID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  HAVING AVG(R.Rating) = (
 10      SELECT MAX(Average_Rating)
 11      FROM (
 12          SELECT AVG(Rating) AS Average_Rating
 13          FROM Review
 14          GROUP BY ProductID
 15      )
 16  );

PRODUCT PRODUCT_NAME         AVERAGE_RATING
------- -------------------- --------------
P000004 Lipstick                       4.67
```

4.FIND PRODUCTS WITH AVERAGE RATING ABOVE 4

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      ROUND(AVG(R.Rating), 2) AS Average_Rating
  5  FROM Product P
  6  JOIN Review R
  7  ON P.Product_ID = R.ProductID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  HAVING AVG(R.Rating) > 4
 10  ORDER BY Average_Rating DESC;

PRODUCT PRODUCT_NAME         AVERAGE_RATING
------- -------------------- --------------
P000004 Lipstick                       4.67
```

6.IDENTIFY HIGHLY RATED PRODUCTS

GENERATE PRODUCT RATING REPORTS

REPORT 1 : PRODUCT RATING ANALYSIS

DISPLAY:

- PRODUCT NAME
- NUMBER OF REVIEWS
- AVERAGE RATING

```
SQL> SELECT
  2      P.Product_Name,
  3      COUNT(R.ReviewID) AS Number_Of_Reviews,
  4      ROUND(AVG(R.Rating), 2) AS Average_Rating
  5  FROM Product P
  6  LEFT JOIN Review R
  7  ON P.Product_ID = R.ProductID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Average_Rating DESC NULLS LAST;

PRODUCT_NAME         NUMBER_OF_REVIEWS AVERAGE_RATING
-------------------- ----------------- --------------
Lipstick                             3           4.67
Laptop                               2              4
Face Wash                            2            3.5
Face Cream                           5            3.2
Kids Shoes                           2              3

SQL>
```

Report 2: Customer Feedback Analysis

A. Most reviewed products

```
SQL> SELECT
  2      P.Product_Name,
  3      COUNT(R.ReviewID) AS Review_Count
  4  FROM Product P
  5  JOIN Review R
  6  ON P.Product_ID = R.ProductID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  ORDER BY Review_Count DESC;

PRODUCT_NAME         REVIEW_COUNT
-------------------- ------------
Face Cream                      5
Lipstick                        3
Laptop                          2
Kids Shoes                      2
Face Wash                       2
```

B. Highly rated products

```
SQL> SELECT
  2      P.Product_Name,
  3      ROUND(AVG(R.Rating), 2) AS Average_Rating
  4  FROM Product P
  5  JOIN Review R
  6  ON P.Product_ID = R.ProductID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  HAVING AVG(R.Rating) >= 4.5
  9  ORDER BY Average_Rating DESC;

PRODUCT_NAME         AVERAGE_RATING
-------------------- --------------
Lipstick                       4.67
```

C. Products requiring improvement

```
SQL> SELECT
  2      P.Product_Name,
  3      COUNT(R.ReviewID) AS Review_Count,
  4      ROUND(AVG(R.Rating), 2) AS Average_Rating
  5  FROM Product P
  6  JOIN Review R
  7  ON P.Product_ID = R.ProductID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  HAVING AVG(R.Rating) < 3
 10  ORDER BY Average_Rating;

no rows selected
```

Report 3: Rating Distribution

A. Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 5;

FIVE_STAR_RATINGS
-----------------
                5
```

B. Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS Four_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 4;

FOUR_STAR_RATINGS
-----------------
                3
```

C. Number of low-rated products

```
SQL> SELECT COUNT(*) AS Low_Rated_Products
  2  FROM (
  3       SELECT ProductID
  4       FROM Review
  5       GROUP BY ProductID
  6       HAVING AVG(Rating) < 3
  7  );

LOW_RATED_PRODUCTS
------------------
                 0
```

D. Complete rating distribution

```
SQL> SELECT
  2      Rating,
  3      COUNT(*) AS Number_Of_Ratings
  4  FROM Review
  5  GROUP BY Rating
  6  ORDER BY Rating DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 5
         4                 3
         3                 3
         2                 2
         1                 1
```